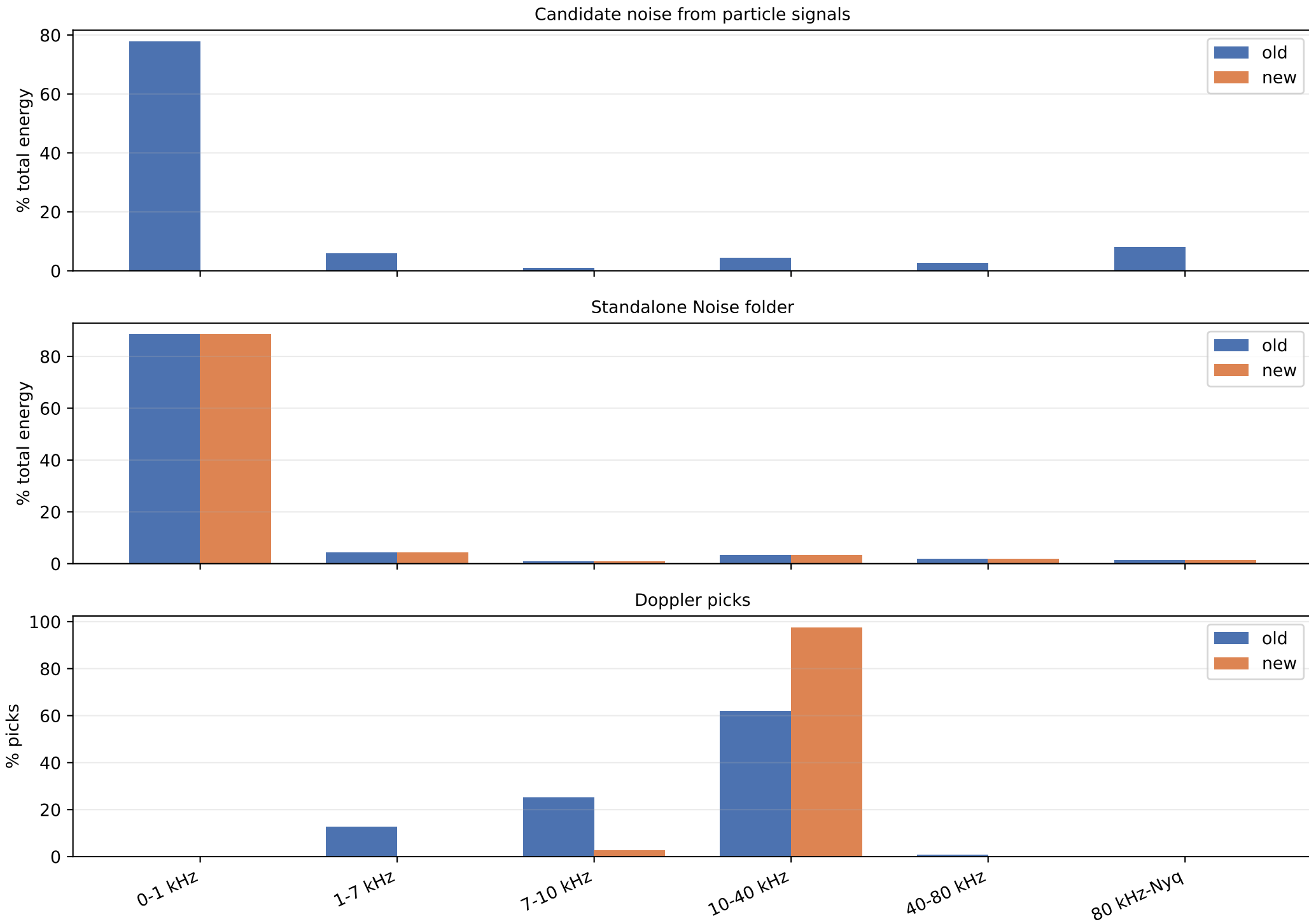


# Spectral Noise vs Doppler Picks - Old/New Pipeline

Candidate noise = windows from particle signals that do not overlap detected particles ( $t_0 \pm 3 \tau$ ).  
Standalone noise = files from the Noise folder, analyzed with the same class FFT settings.  
Interpretation: shared dominant bands are consistent with contamination by weak or missed events.

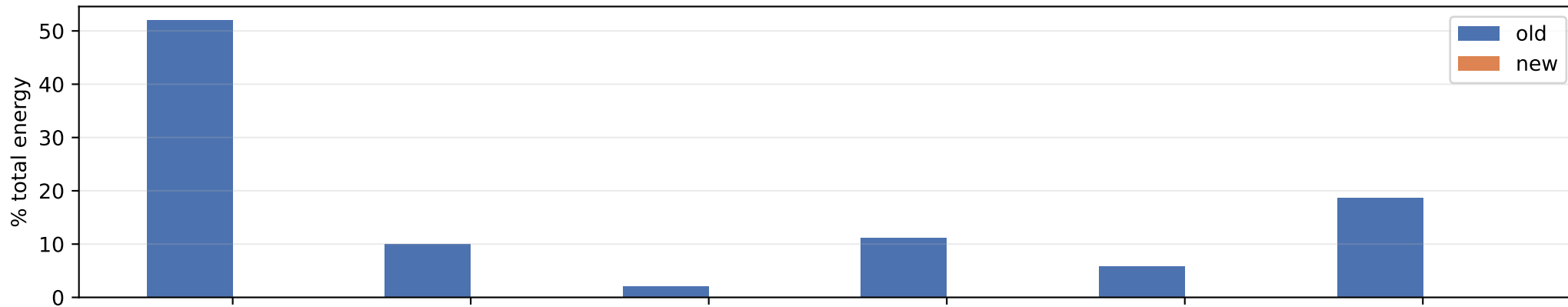
```
old 2um: candidate 10-40 kHz=4.5%, Noise=3.3%, Doppler picks=61.8%, dominant overlap=False
old 4um: candidate 10-40 kHz=11.2%, Noise=3.2%, Doppler picks=74.2%, dominant overlap=False
old 10um: candidate 10-40 kHz=0.4%, Noise=3.2%, Doppler picks=61.1%, dominant overlap=False
new 2um: candidate 10-40 kHz=nan%, Noise=3.3%, Doppler picks=97.5%, dominant overlap=False
new 4um: candidate 10-40 kHz=nan%, Noise=3.2%, Doppler picks=100.0%, dominant overlap=False
new 10um: candidate 10-40 kHz=nan%, Noise=3.2%, Doppler picks=97.6%, dominant overlap=False
```

## 2um - Old vs New Spectral Comparison

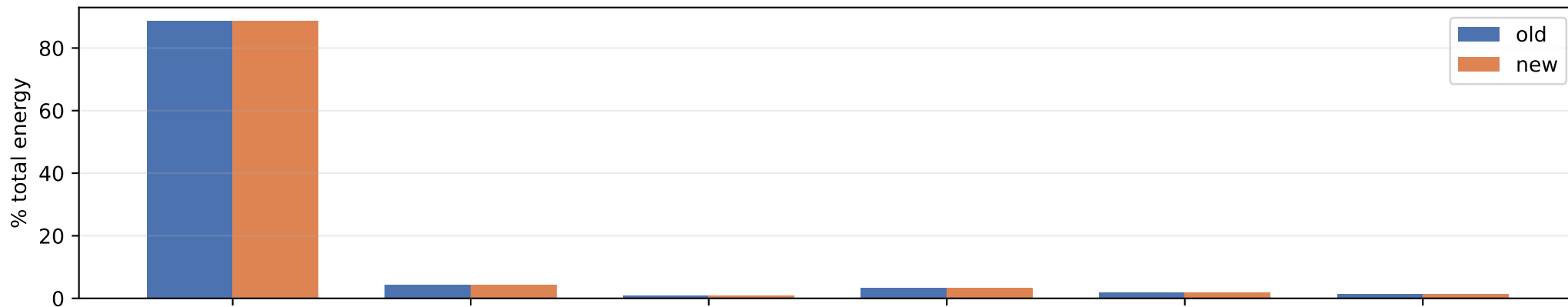


# 4um - Old vs New Spectral Comparison

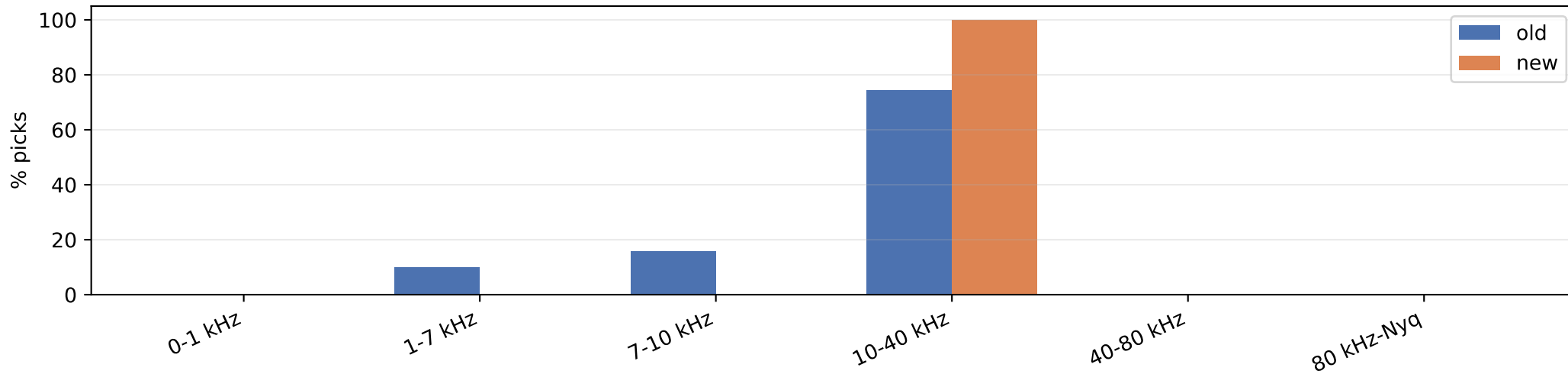
Candidate noise from particle signals



Standalone Noise folder

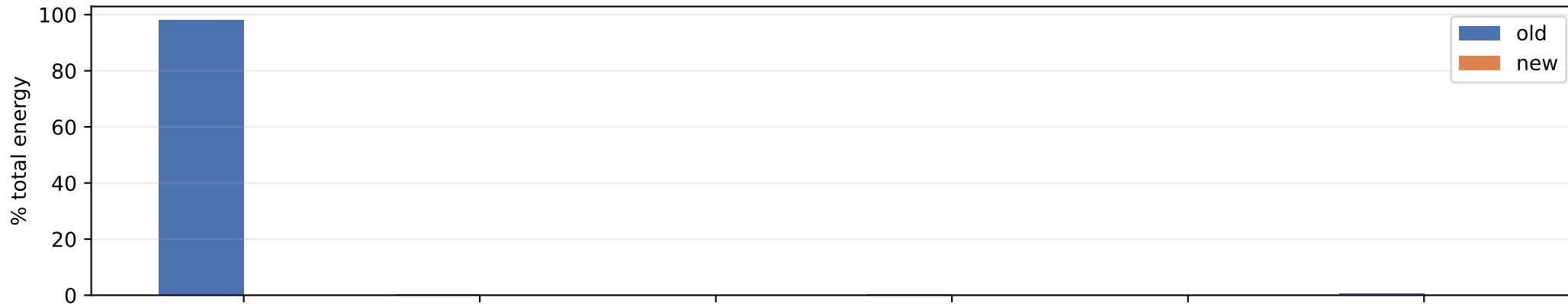


Doppler picks

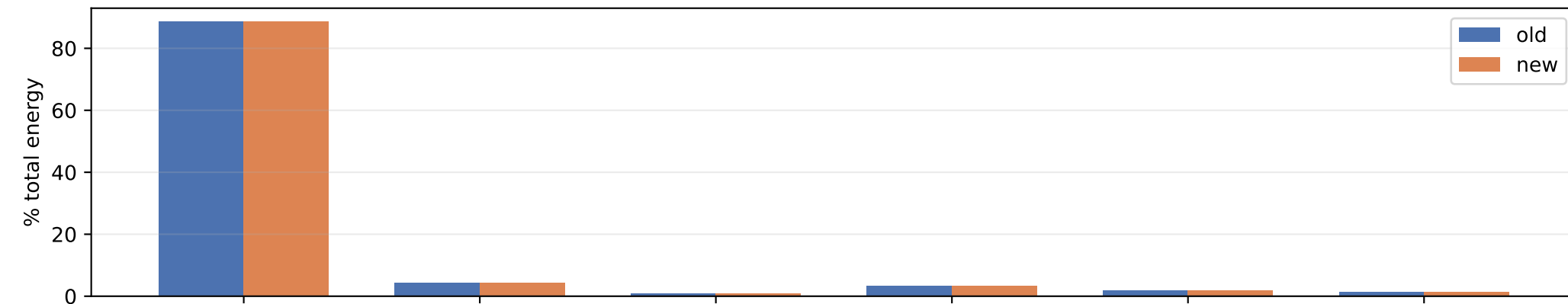


# 10um - Old vs New Spectral Comparison

Candidate noise from particle signals



Standalone Noise folder



Doppler picks

